Air Quality Visualization and Forecasting of Air Quality Index Using a Hybrid Machine Learning Model (Random Forest, XGBoost, LSTM with ARIMA Residual Modelling)

A thesis submitted in partial fulfilment of the requirements

for the award of the degree of

Master of Science in

Mathematics with Data Science

By

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Roll Number-2445005

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SCHOOL OF APPLIED SCIENCES

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

DEEMED TO BE UNIVERSITY

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DECLARATION

I hereby declare that the work reported in M.sc thesis **entitled Air Quality Visualization and Forecasting of Air Quality Index Using a Hybrid Machine Learning Model (Random Forest, XGBoost, LSTM with ARIMA Residual Modelling)** is submitted in the **School of Applied Sciences, KIIT Deemed to be University, Bhubaneswar**, is a Bonafide record of my work carried out under the supervision of **Dr. Vishal Pradhan** have not submitted this work elsewhere for any other degree or diploma.

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CERTIFICATE

This is to certify that the thesis titled **Air Quality Visualization and Forecasting of Air Quality Index Using a Hybrid Machine Learning Model (Random Forest, XGBoost, LSTM with ARIMA Residual Modelling)** is the work done by **Priyanka (2445005)** in partial fulfilment of the requirements for the award of the degree of **Master of Science in Mathematics with Data Science of the School of Applied Sciences, KIIT Deemed to be University, Bhubaneswar under my/our supervision and this work has not been submitted elsewhere for a degree.**

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Thesis Viva-voce held on _____

Internal Examiner

External Examiner

ABSTRACT

Air pollution is a major environmental and public health concern, especially in rapidly urbanizing regions. This study focuses on analyzing and forecasting air quality using a data-driven approach. The datasets, obtained from Kaggle, include **city_hour.csv** (hourly pollution data for cities such as Delhi and Patna) and **stations.csv** (monitoring station metadata).

A hybrid modeling approach is proposed, combining **XGBoost, Random Forest, and LSTM** for AQI prediction, with **ARIMA** applied to model residuals. This integration improves forecasting accuracy by leveraging both machine learning and time-series techniques. Model interpretability is enhanced using SHAP values to understand the contribution of each pollutant.

The study utilizes key pollutants such as PM2.5, PM10, NO_2, SO_2, CO, and O_3, along with meteorological factors like humidity, wind speed, and pressure. Data preprocessing, including cleaning and handling missing values through linear interpolation, ensures data consistency and reliability.

Exploratory Data Analysis (EDA) is conducted to identify trends and relationships. AQI is computed using the CPCB standard sub-index method, where the final AQI is determined as the maximum of individual pollutant indices.

The models are used to forecast pollutant levels over a long-term period, and AQI is estimated based on these predictions. ARIMA further refines results by capturing residual patterns. The findings demonstrate that hybrid models effectively capture complex pollution trends, though long-term predictions remain influenced by external environmental factors.

Overall, this study provides a structured framework for air quality analysis and forecasting, offering insights for environmental monitoring and policy planning.

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CHAPTER -1

INTRODUCTION

The Need for Decentralized and Sustainable Air Quality Forecasting in Smart Cities

1. Introduction Rapid urbanization and industrialization have drastically increased air pollution, making accurate Air Quality Index (AQI) forecasting essential for smart city management, public health, and environmental planning. Because emissions cause localized air quality variations [1], city authorities increasingly require fine spatial and temporal data resolution [2]. Long-term exposure to airborne pollutants is strongly linked to respiratory and cardiovascular diseases [3], [4], [5], making predictive models vital for early health warnings and risk mitigation [6].

2. Limitations of Traditional Centralized Models Traditional machine learning (ML) and deep learning (DL) models [7], [8], [9] use historical data to link contaminants to AQI levels. Yet, these centralized frameworks suffer from several systemic flaws:

- Security and Privacy: Relying on centralized data aggregation is resource-intensive and inherently compromises data privacy [7], [8], [9], [10], [11].
- Lack of Spatial Awareness: Traditional methods underestimate the varied toxicity of pollutants across regions and fail during extreme pollution events [12]. Their lack of spatial distribution awareness limits their utility in metropolitan areas [7], [8], [9], [10].
- Poor Sustainability: Sustainable urban environments rely on optimal energy and power consumption metrics [13]. Traditional models [7], [8], [9], [10] are environmentally harmful due to their high energy demands and assumption of centralized infrastructure, which makes them poorly suited for the fragmented data flows of modern cities.

3. The Case for Decentralized Learning in Urban Environments Urban air quality monitoring is naturally decentralized, relying on distributed sensing nodes managed by various agencies. Centralized learning is operationally inefficient in these large-scale deployments due to several system-level constraints:

- High Communication Energy Costs: Centralized systems demand continuous, high-resolution data transmission to the cloud. This sustained uplink requirement creates an energy bottleneck for battery- or solar-powered sensors, as communication energy often dominates computation energy [14].
- Processing Latency: Cloud-side data aggregation and retraining introduce significant end-to-end delays. In contrast, edge-level training (such as federated

learning) bypasses round-trip cloud dependencies, enabling the real-time predictions needed for public alerts and traffic regulation.

- Scalability Bottlenecks: As sensor networks grow, centralized models face data ingestion bottlenecks and single-point-of-failure risks. Federated learning, however, scales horizontally [15], easily accommodating the addition or removal of urban sensors without breaking the global data pipeline.

Bridging the Data Gap: Decentralized Solutions for Rural and Citizen-Centric AQI

1. The Rural Data Desert and the Accessibility Barrier While decentralized learning is architecturally superior for urban hubs, its implementation is often hindered by the high cost of regulatory-grade monitoring stations, which remain concentrated in industrial sectors or weather agencies. Common citizens and rural populations frequently lack access to "air quality machines," creating a significant data void in non-urban areas. To address this, the solution lies in a multi-tiered sensing ecosystem that moves beyond agency-owned infrastructure to include low-cost, community-hosted hardware [16].

2. Federated Citizen Science and Low-Cost IoT The primary strategy for rural inclusion is the deployment of low-cost, consumer-grade Internet of Things (IoT) sensors. While individual low-cost sensors are less precise than industrial stations, a dense network of these devices—leveraged through Federated Learning (FL)—can achieve high collective accuracy [17]. In this model, decentralized learning allows for local "noise" reduction and calibration, enabling the global model to learn from thousands of "good enough" home-based nodes without requiring rural residents to own expensive laboratory equipment [18].

3. Hybrid Satellite-Edge Data Fusion For areas where even low-cost sensors are unavailable, the solution involves fusing decentralized local data with Satellite Remote Sensing. Satellite imagery provides broad atmospheric baselines, while the few available decentralized nodes (such as smartphones or localized community sensors) act as "ground-truth" anchors [19]. This hybrid approach uses FL to fine-tune global satellite models based on specific micro-climates in rural zones, ensuring that even sparse data contributes to high-resolution local forecasting [20].

4. Mobile Sensing and Virtual Infrastructure To further democratize access, the "common person" can contribute via mobile sensing. By mounting sensors on public transit, mail delivery vehicles, or agricultural machinery, rural areas can be mapped dynamically without stationary infrastructure [21].

CHAPTER 2

LITERATURE REVIEW

Air pollution has emerged as a critical environmental and public health issue worldwide. **The Air Quality Index (AQI)** is widely used to quantify air pollution levels by integrating multiple pollutants such as **PM2.5, PM10, NO_2, SO_2, CO, and O_3.** Accurate prediction of AQI is essential for environmental monitoring, risk assessment, and policy formulation.

Traditional Time-Series Models

Statistical models such as the **Auto Regressive Integrated Moving Average (ARIMA)** have been extensively used for **AQI forecasting. ARIMA** effectively captures linear trends and temporal dependencies in time-series data **[23](Box et al., 2015).** However, it is limited in modelling nonlinear relationships among pollutants.

Machine Learning Models

Machine learning models such as Random Forest and XGBoost have gained popularity in AQI prediction due to their ability to model nonlinear relationships. **Random Forest [24](Breiman, 2001)** is an ensemble method that improves accuracy by aggregating multiple decision trees**, while XGBoost [25] (Chen & Guestrin, 2016)** enhances performance through gradient boosting and regularization. Although these models provide high predictive accuracy, they lack the ability to explicitly capture temporal dependencies in sequential data.

Deep Learning Models

Deep learning techniques, particularly Long Short-Term Memory (LSTM) networks, are effective for time-series forecasting**. LSTM** models can capture long-term dependencies and temporal patterns in AQI data [26] **(Hochreiter & Schmidhuber, 1997).** However, these models may still leave residual errors due to unmodeled linear components.

Ensemble and Hybrid Approaches

To overcome the limitations of individual models, several researchers have proposed ensemble and hybrid approaches for AQI prediction. For instance, studies by [27] **Zhang et al. (2018) and [28] Li et al. (2020)** demonstrated that combining machine learning models such as Random Forest and gradient boosting techniques significantly improves prediction accuracy by capturing complex nonlinear relationships among pollutants. Similarly, deep learning models like **LSTM** have been widely used to model temporal dependencies in air quality data, as shown in the work of **Qin et al. (2019), where LSTM effectively captured long-term sequential patterns in AQI time series.**

Building on these approaches, recent research has focused on hybrid frameworks that integrate statistical and machine learning models. For example, **Wang et al. (2021)** proposed a hybrid model where machine learning predictions were further refined using

ARIMA applied to residual errors. In such frameworks, the ensemble model captures the dominant nonlinear and temporal structures in the data, while **ARIMA models the remaining linear components present in the residuals.** This two-stage modelling approach has been shown to reduce prediction errors and enhance robustness, making it highly suitable for complex environmental datasets.

Model Interpretability using SHAP

With the increasing complexity of machine learning models, interpretability has become a crucial aspect of **AQI prediction. Shapley Additive explanations (SHAP), introduced by Lundberg and Lee (2017), is a game-theoretic approach used to explain model predictions.**

SHAP assigns an importance value to each feature by measuring its contribution to the prediction**. In AQI prediction, SHAP helps identify the influence of individual pollutants such as PM2.5, PM10, NO_2, SO_2, CO, and O_3 on the predicted AQI values.**

The use of **SHAP** with models such as **XGBoost** provides both global interpretability **(overall feature importance)** and local interpretability **(individual prediction explanation)**. This enhances transparency and reliability, making the model more suitable for environmental decision-making and policy analysis**.**

Data Preprocessing Techniques

Data preprocessing is essential for improving model performance. **Missing values in AQI datasets** are commonly handled using **interpolation techniques. Linear interpolation** is preferred **due to its ability to maintain temporal continuity without introducing artificial fluctuations.**

Normalization techniques such as Min-Max scaling are applied to ensure that all features are on a comparable scale, improving model convergence and stability.

Research Gap

Despite advancements in AQI prediction, existing studies often focus on individual models or limited hybrid approaches. There is a lack of comprehensive frameworks that integrate multiple machine learning models with residual-based statistical correction and interpretability techniques**.**

Conclusion

Based on the reviewed literature, combining Random Forest, XGBoost, and LSTM provides a robust framework for capturing nonlinear and temporal patterns in AQI data. The application of ARIMA on residuals further enhances prediction accuracy. Additionally, incorporating SHAP improves model interpretability by identifying the contribution of individual pollutants.

CHAPTER 3

PROBLEM FORMULATION AND METHODLOGY

Problem Formulation

Air pollution has become a major environmental and health concern , and the Air Quality Index is used to measure the level of pollution based on various pollutants such as **PM2.5, PM10, NO_2, SO_2, CO, and Ozone.**

The problem arises in Predicting AQI due to nonlinear relationships and temporal dependencies in the data . Traditional models like ARIMA capture linear patterns, while machine learning models such as Random Forest and XGBoost model nonlinear relationships but do not inherently capture time-dependent patterns. LSTM networks handle temporal dependencies but may leave residual errors can limit the overall performance of the forecasting system.

Another challenge is that AQI datasets often contain missing values due to sensor malfunction or data recording issues. Proper preprocessing techniques, such as linear interpolation and normalization, are necessary to ensure the reliability of the data. Furthermore, many advanced machine learning models act as black boxes, making it difficult to understand how each pollutant contributes to the predicted AQI. This lack of interpretability can reduce the trust of policymakers and environmental agencies.

Considering these challenges, there is a need for a comprehensive and interpretable framework that can accurately forecast **AQI**. Therefore, this study proposes a hybrid modeling approach that combines **Random Forest, XGBoost, and LSTM** to capture both nonlinear and temporal patterns in the data. The residual errors from the combined model are further modelled using **ARIMA** to enhance prediction accuracy. In addition, **Shapley Additive explanations (SHAP)** are employed to interpret the contribution of each pollutant to the AQI predictions.

To solve this problem, the following steps are undertaken:

*** Data preprocessing: it includes handling missing values using linear interpolation and to improve machine learning model performance. Common methods include Min-Max scaling.**

*** Data Modelling : we constructed a hybrid ensemble- model by combining predictions from RF, XGBoost, and LSTM to obtain a AQI forecasting.**

*** Data Residual : The modelled AQI has errors of the ensemble prediction using ARIMA to capture remaining linear temporal patterns.**

*** Pollutant interpretability :** Shapley Additive Explanations (SHAP) for identifying the contribution of individual pollutants.

*** Data Analytics: To evaluate model performance using statistical metrics such as MAE, RMSE, MAPE, and R^2.**

*** Visualization AQI and Residuals: To visualize AQI trends and forecasting results for improved understanding and decision support. Plot error created by forecasting as residual comparison with Arima-residuals**

METHODOLOGY:

Air Quality datasets often contain missing observations due to sensor malfunction or data recording issues. To ensure temporal continuity and data integrity, linear interpolation is employed for HANDLING MISSING VALUES, This method provides smooth transitions between known data points without introducing artificial fluctuations. Then , we try to extract the best features by CORRELATION AND HEATMAPS ,there the selected input features are pollutant concentrations that significantly influence AQI. And also the factors affecting pollutants. Now, when we deploy models specifically LSTM there we use Data Normalization to ensure uniform feature scaling and improve model convergence, Min-Max normalization is applied only on the training portion to prevent data leakage. After ,scaling the training data we split the TRAIN-TEST data according to x-scaler and y-scaler . Hence , The dataset is divided into 70% training data and 30% testing data while preserving temporal order to avoid data leakage**.**

Model Development

For further analysis, the original pollutant concentrations were not directly aligned with the Central Pollution Control Board (CPCB) breakpoint standards. Therefore, sub-indices for each pollutant were first computed, and the CPCB-approved AQI formulation was subsequently applied to derive standardized AQI values.

To enable long-term forecasting, AQI predictions were generated for an eight-year horizon. Initially, the concentrations of individual pollutants were forecasted using the Extreme Gradient Boosting (XGBoost) algorithm, which minimizes prediction errors through a regularized objective function and offers high predictive accuracy along with computational efficiency. The predicted pollutant values were then utilized as inputs for AQI prediction using XGBoost, Random Forest (RF), and Long Short-Term Memory (LSTM) models. Since the predicted AQI values did not always conform to CPCB-defined categories, sub-indices were recalculated using the specified CPCB breakpoints to obtain the standardized AQI_CPCB values.

Subsequently, a residual distribution analysis was performed using an ensemble strategy that leverages the strengths of individual models. The predictions from RF, XGBoost, and LSTM were combined using a simple averaging technique to produce a single ensemble estimate. While RF and XGBoost effectively captured nonlinear relationships among pollutant concentrations, the LSTM model accounted for temporal dependencies inherent in the time-series data, thereby reducing individual model biases and variances.

Despite the enhanced performance achieved through ensemble, systematic patterns persisted in the prediction errors. To address this, residuals were computed as the difference between the actual AQI values and the ensemble predictions. These residuals were modeled using the Auto-Regressive Integrated Moving Average (ARIMA) approach, which is well-suited for capturing remaining linear temporal dependencies. The optimal

ARIMA parameters (p, d, q)were determined using the Auto-ARIMA method based on the Akaike Information Criterion (AIC). The predicted residuals were subsequently added to the ensemble estimates to obtain the final hybrid AQI forecasts, thereby improving overall predictive accuracy.

The performance of the proposed hybrid framework was evaluated using standard statistical metrics, including Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), and the Coefficient of Determination (R^2), ensuring a comprehensive assessment of both accuracy and reliability.

Furthermore, to enhance model transparency and interpretability, SHapley Additive exPlanations (SHAP) were applied, particularly to the XGBoost component, to quantify the contribution of each pollutant—PM2.5, PM10, NO_2, SO_2, CO, and O_3—to the predicted AQI values. Finally, various visualization techniques, including time-series plots of actual versus predicted AQI, residual plots with a zero-residual baseline, and SHAP summary and dependence plots, were employed to facilitate intuitive interpretation and effective communication of the results.

CHAPTER 4

DATA PREPROCESSING

Data preprocessing is a crucial step in the development of any machine learning and time-series forecasting model. Real-world air quality datasets often contain inconsistencies such as missing values, varying scales among features, and temporal misalignments. If not properly addressed, these issues can significantly affect the accuracy and reliability of predictive models. Therefore, prior to model development, the dataset was carefully pre-processed to ensure data quality and consistency. **In this study, three primary preprocessing techniques were employed: handling missing values using linear interpolation, feature scaling through Min–Max normalization, and temporal data alignment and feature selection.**

I . Handling Missing Values

a. Need for Handling Missing Data

Air quality monitoring systems frequently experience data gaps due to sensor malfunctions, maintenance activities, or transmission errors. Missing values can lead to biased model training and reduced predictive performance. Hence, it is essential to impute these missing observations in a manner that preserves the natural temporal structure of the data.

b. Linear Interpolation Method

In this study, linear interpolation was used to estimate missing values. This method assumes that the change between two known data points is linear and fills the missing values accordingly. It is particularly suitable for environmental time-series data where pollutant concentrations typically change gradually over time.

c. Mathematical Representation

If x_t is a missing value between two known observations x_{t_1} and x_{t_2}, the interpolated value is calculated as:

$$x_t = x_{t_1} + \frac{(x_{t_2} - x_{t_1})}{(t_2 - t_1)} \times (t - t_1)$$

d. Advantages

- Preserves temporal continuity.
- Simple and computationally efficient.
- Avoids introduction of artificial noise.
- Suitable for smoothly varying environmental data.

II. Feature Selection

Selecting relevant features helps improve model performance, reduce computational complexity, and enhance interpretability. In AQI prediction, pollutant concentrations are the primary determinants of air quality index. We also compare correlation between pollutants and the its effected factors.

Selected Features

The following pollutants were selected as input variables based on their significance in AQI computation:

- PM2.5 (µg/m³)
- PM10 (µg/m³)
- NO_2 (µg/m³)
- SO_2 (µg/m³)
- CO (mg/m³)
- O_3 (µg/m³)

Effected factors:

- Benzene (ug/m3)
- Toluene (ug/m3)
- RH (%)
- WS (m/s)
- WD (degree)
- SR (W/mt2)
- BP (mmHg)
- AT (degree C)

The target variable for prediction is the Final AQI (AQI_FINAL)

The factors effects are plotted via heatmap for the feature selection as data is temporal the chances og getting a strong correlation is very less.

Advantages

- Reduces model complexity.
- Enhances prediction accuracy.
- Improves interpretability, especially when using SHAP analysis.

III. Feature Scaling using Min-Max Normalization

a. Need for Scaling

The pollutant variables in the dataset are measured in different units and ranges (e.g., PM2.5 in µg/m³ and CO in mg/m³). Models such as LSTM are sensitive to the scale of input features, and unscaled data can lead to slow convergence and suboptimal performance. Therefore, scaling is necessary to ensure that all features contribute equally to the model training process.

b. Min–Max Normalization

Min–Max scaling transforms the original data into a fixed range, typically [0, 1], without distorting the relationships among features.

c. Mathematical Representation

$$\boldsymbol{x_{scaled}} = \frac{\boldsymbol{x - x_{min}}}{\boldsymbol{x_{max} - x_{min}}}$$

where:

- $X$is the original value,
- $X_{\min}$and $X_{\max}$ are the minimum and maximum values of the feature.

d. Advantages

- Preserves the original distribution of the data.
- Enhances convergence of neural networks.
- Ensures equal contribution of all features.
- Improves numerical stability.

CHAPTER 5

MODEL DEPLOYMENT AND COMPUTATONAL ANALYSIS

5.1 EVALUATION METRIC:

5.1.1 MEAN SQUARED ERROR (MSE)

The Mean Squared Error (MSE) is utilized to quantify the average squared difference between the actual and predicted AQI values. It penalizes larger errors more heavily, making it particularly useful for assessing the precision of regression models. A lower MSE indicates better model performance and higher predictive accuracy.

Mathematical Formula

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

Interpretation

MSE Value	Interpretation
Close to 0	Excellent predictive accuracy
Moderate	Acceptable performance
High	Poor model performance

5.1.2 ROOT MEAN SQUARED ERROR (RMSE)

Definition

Root Mean Squared Error (RMSE) is a widely used statistical metric for evaluating the accuracy of regression models. It measures the square root of the average of the squared differences between the actual observed values and the predicted values. RMSE provides an estimate of the magnitude of prediction errors in the same units as the target variable, making it highly interpretable.

Mathematical Formula

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

Where:

- n= **Number of observations**
- y_i= **Actual (observed) value**
- $\hat{y}_i$= **Predicted value**
- $(y_i - \hat{y}_i)$= **Prediction error (residual)**

Relationship with MSE

RMSE is directly derived from the Mean Squared Error (MSE):

$$\text{RMSE} = \sqrt{\text{MSE}}$$

- MSE expresses error in squared units.
- RMSE converts this error back to the original units of the dependent variable (e.g., AQI), making it easier to interpret.

5.1.3 COEFFICIENT OF DETERMINATION (R^2)

The Coefficient of Determination (R^2) measures the proportion of variance in the observed AQI values that is explained by the predictive model. It provides an indication of the model's goodness-of-fit, with values closer to 1 signifying superior explanatory power.

Mathematical Formula

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

Interpretation

R^2 Value	Interpretation
1	Perfect fit
0.7 – 0.9	Strong predictive capability

R^2 Value	Interpretation
0.5 – 0.7	**Moderate fit**
< 0.5	**Weak model performance**
Negative	**Model performs worse than the mean**

5.1.4 ADF TEST FOR STATIONARITY:

The Augmented Dickey-Fuller (ADF) test is a formal statistical test used to check for the presence of a unit root in a time series **(An & Zhao, 2017).**

- **Stationary Series:** The statistical properties (mean, variance) do not change over time. These are easier to predict.
- **Non-Stationary Series:** These exhibit trends or seasonal patterns, making them unpredictable unless transformed (Wang et al., 2020).

How it works:

The test evaluates a null hypothesis (H_0) that the data is non-stationary.

- **If the p-value < 0.05, you reject the null hypothesis; the data is stationary.**
- **If the p-value > 0.05, you fail to reject the null; the data is non-stationary (An & Zhao, 2017).**

DEFINE

*** ADF Statistics: This is the "Test Statistic." It measures how strongly the data depends on a trend. The more negative this number is, the more likely the data is stationary (stable).**

*** p-value: This represents the probability that your data has a unit root (is non-stationary). In statistics, we typically look for a value below 0.05 to claim success.**

*** Critical Levels (1%, 5%, 10%): these are "benchmarks" or hurdles. To prove your data is stationary, your ADF Statistic must be smaller (more negative) than these values.**

- **1%: High certainty.**
- **5%: Standard scientific certainty.**
- **10%: Low certainty.**

Table 1: TABLE FOR ADF TEST (*STATIONARITY TEST FOR POLLUTANTS*) [DL030]:

ADF Statistics	**-17.647**
p-value	3.734545571876759e-30
1% critical level	**-3.43**
5% critical level	**-2.86**
10% critical level	**-2.566**

Table 2: TABLE FOR ADF TEST (*STATIONARITY TEST FOR POLLUTANTS*) [DL024]:

ADF Statistics	**-19.293**
p-value	**0.0**
1% critical level	**-3.431**
5% critical level	**-2.861**
10% critical level	**-2.566**

Table 3: TABLE FOR ADF TEST (*STATIONARITY TEST FOR POLLUTANTS*) [BR005]:

ADF Statistics	**-12.167**
p-value	1.4435857035609038e-22
1% critical level	**-3.43**
5% critical level	**-2.86**
10% critical level	**-2.566**

INTERPRETATION:

FIG[1] ADF statistic is -17.64 less than all critical levels , then result is highly significant. The p value is 3.73 *10^-30 which is very very small near to 0 which means the dataset is **STATIONARY.**

FIG[2] ADF statistics is -19.293 less than all critical levels, then result is highly significant . The p value is 0 which is 0 which means the dataset is **STATIONARY.**

FIG[3] ADF statistics is -12.005 less than all critical levels , , then result is highly significant. The p value is 1.44 *10^-22 which is very very small near to 0 which means the dataset is **STATIONARY.**

5.2.5 POLLUTANTS PERFORMANCE METRIC FOR BR005:

XGBoost Pollutant Prediction Metrics	MSE	RMSE	R^2
PM 2.5 (ug/m3)	14.03	3.75	0.89
PM 10 (ug/m3)	26.12	5.11	0.87
NO2 (ug/m3)	8.04	2.84	0.97
SO2 (ug/m3)	2.22	1.49	0.79
CO (mg/m3)	0.22	0.34	0.84
OZONE (ug/m3)	7.84	2.80	0.84

*** Pollutant-Wise Interpretation**

PM2.5 (µg/m³)

- **R^2 = 0.89 indicates strong predictive performance.**
- **RMSE = 3.75 suggests moderate prediction error.**
- **The model effectively captures variations in fine particulate matter.**

PM10 (µg/m³)

- **R^2 = 0.87 reflects good predictive capability.**
- **Highest RMSE (5.11) indicates greater variability and complexity in coarse particulate matter.**

NO_2 (µg/m³)

- **R^2 = 0.97 signifies excellent model performance.**
- **Low RMSE (2.84) indicates highly accurate predictions.**
- **Suggests stable and predictable emission patterns.**

SO_2 (µg/m³)

- **R^2 = 0.79 indicates moderate predictive strength.**
- **Very low RMSE (1.49) suggests small absolute errors despite slightly lower variance explanation.**

CO (mg/m³)

- **R^2 = 0.84 demonstrates good predictive accuracy.**
- **RMSE = 0.47 is small due to the lower concentration scale of CO.**

O_3 (µg/m³)

- **R^2 = 0.84 indicates good model performance.**
- **RMSE = 2.80 reflects reasonable prediction accuracy for this secondary pollutant.**

5.2.5 POLLUTANT PERFORMANCE METRIC FOR DL024:

XGBoost Pollutant Prediction Metrics	MSE	RMSE	R^2
PM 2.5 (ug/m3)	31.20	5.59	0.83
PM 10 (ug/m3)	60.73	7.79	0.77
NO2 (ug/m3)	0.00	0.00	1.00
SO2 (ug/m3)	2.45	1.57	0.75
CO (mg/m3)	0.41	0.64	0.70
OZONE (ug/m3)	9.15	3.02	0.60

Pollutant-Wise Analysis

PM2.5 (µg/m³)

- **R^2 = 0.83 indicates strong predictive capability.**
- **RMSE = 5.59 suggests moderate prediction error.**
- **As a primary contributor to AQI, accurate prediction is crucial for reliable AQI forecasting.**

PM10 (µg/m³)

- **R^2 = 0.77 reflects good but comparatively lower predictive strength.**
- **Highest RMSE (7.79) indicates significant variability due to sources like dust and construction activities.**

NO_2 (µg/m³)

- **R^2 = 1.00 and MSE = 0 suggest a perfect fit.**
- **Such perfection is rare and may indicate:**
    - **Data leakage.**
    - **Overfitting.**
    - **Extremely low variability in the dataset.**
- **It is advisable to verify the data and model evaluation process.**

SO_2 (µg/m³)

- **R^2 = 0.75 indicates moderate predictive performance.**
- **Low RMSE (1.57) suggests that absolute errors are small.**

CO (mg/m³)

- **R^2 = 0.70 reflects acceptable predictive capability.**
- **RMSE = 0.64 is reasonable given the smaller concentration scale.**

O_3 (µg/m³)

- **R^2 = 0.60 indicates the lowest predictive performance among the pollutants.**
- **Ozone formation depends heavily on meteorological and photochemical processes, making it more challenging to model.**

5.2.6 POLLUTANT PERFORMANCE METRIC FOR DL030:

XGBoost Pollutant Prediction Metrics	MSE	RMSE	R^2
PM 2.5 (ug/m3)	27.08	41.67	0.82
PM 10 (ug/m3)	66.63	94.28	0.73
NO2 (ug/m3)	8.84	13.58	0.74
SO2 (ug/m3)	2.84	5.01	0.78
CO (mg/m3)	0.41	0.66	0.71
OZONE (ug/m3)	10.50	15.81	0.81

Pollutant-Wise Performance Analysis

PM2.5 (µg/m³)

- R^2 = 0.82 indicates strong predictive capability.
- Correct RMSE ≈ 5.20 µg/m³, suggesting moderate prediction error.
- As a major contributor to AQI, accurate prediction is crucial.

PM10 (µg/m³)

- R^2 = 0.73 reflects moderate predictive performance.
- Highest RMSE, indicating significant variability due to sources like dust and construction.

NO_2 (µg/m³)

- R^2 = 0.74 indicates reasonable predictive strength.
- Moderate RMSE suggests the model captures emission-related variations fairly well.

SO_2 (µg/m³)

- R^2 = 0.78 shows good predictive capability.
- Low RMSE indicates minimal absolute prediction errors.

CO (mg/m³)

- $R^2 = 0.71$ suggests acceptable performance.
- **RMSE is small due to the lower concentration scale of CO.**

O_3 (µg/m³)

- $R^2 = 0.81$ indicates good predictive performance.
- Ozone formation is influenced by meteorological and photochemical processes, making accurate prediction noteworthy.

5.3 PSEUDOCODE AND FLOWCHART:

1. Pseudocode for AQI Calculation Using CPCB Guidelines

Objective

To compute the Air Quality Index (AQI) based on pollutant sub-indices defined by the Central Pollution Control Board (CPCB), India.

CPCB AQI Formula

$$I_p = \frac{I_{HI} - I_{LO}}{B_{HI} - B_{LO}}(C_p - B_{LO}) + I_{LO}$$

Where:

- $\boldsymbol{I_p}$**: Sub-index of pollutant** $\boldsymbol{p}$
- $\boldsymbol{C_p}$**: Observed pollutant concentration**
- $\boldsymbol{B_{HI}, B_{LO}}$**: Breakpoint concentrations**
- $\boldsymbol{I_{HI}, I_{LO}}$**: Corresponding AQI breakpoints**

PSEUDOCODE

Algorithm: AQI Calculation Using CPCB Standards
Input:
Dataset D with pollutant concentrations:
PM2.5, PM10, NO2, SO2, CO, O3
CPCB breakpoint table for each pollutant
Output:
AQI values and dominant pollutant for each observation

Steps:
1. *For each observation t in dataset D:*
a. For each pollutant p:
i. Retrieve concentration Cp.

ii. *Identify breakpoint interval such that:*
B_LO <= Cp <= B_HI.
iv. *Compute sub-index:*
*Ip = ((I_HI - I_LO) / (B_HI - B_LO)) * (Cp - B_LO) + I_LO*

b. Determine overall AQI:
AQI_t = max(Ip for all pollutants p)

c. Identify dominant pollutant:
Dominant_p = pollutant with maximum Ip
2. Append AQI_t and Dominant_p to dataset.
3. Return dataset with AQI values.

For each pollutant p:
a. Initialize XGBoost regressor with tuned hyperparameters.
b. Train the model using training data.

4. Prediction:
- Use the trained model to predict pollutant concentrations for the test or future period.

5. Evaluation:
- Compute MSE, RMSE, and R^2 between actual and predicted values.

6. Return predicted pollutant values.

2. Pseudocode for AQI Prediction Using XGBoost, Random Forest, and LSTM

Objective

To predict AQI using an ensemble of machine learning and deep learning models, capturing both nonlinear relationships and temporal dependencies**.**

Pseudocode

Algorithm: AQI Prediction Using XGBoost, Random Forest, and LSTM

Input:
Dataset D with pollutant concentrations and computed AQI

Output:
Predicted AQI values from each model and ensemble prediction

Steps:
1. Data Preprocessing:
a. Handle missing values using linear interpolation.
b. Normalize features for LSTM using Min-Max Scaler.
c. Split dataset into training and testing sets.
. Feature and Target Selection:
- Features X: PM2.5, PM10, NO2, SO2, CO, O3
- Target y: AQI
3. Random Forest Model:

a. Initialize Random Forest Regressor.
b. Train the model on training data.
c. Predict AQI on the test set → y_pred_RF.

4. XGBoost Model:
a. Initialize XGB Regressor.
b. Train the model on training data.
c. Predict AQI on the test set → y_pred_XGB.

5. LSTM Model:
a. Scale features and target using Min-Max Scaler.
b. Reshape input to [samples, timesteps, features].
c. Define LSTM architecture:
- LSTM layer
- Dense output layer
d. Train the model on training data.
e. Predict AQI and inverse-scale → y_pred_LSTM.

6. Ensemble Averaging:
For each observation i:
y_pred_ensemble[i] =
(y_pred_RF[i] + y_pred_XGB[i] + y_pred_LSTM[i]) / 3

7. Model Evaluation:
- Compute MSE, RMSE, and R^2 for each model and the ensemble.

8. Return:
- Individual model predictions.
- Final ensemble AQI predictions.

3. Pseudocode for Hybrid AQI Prediction with ARIMA Residuals

Since your study includes ARIMA-based residual modeling, you may also include the following:

Algorithm: Hybrid AQI Prediction with ARIMA Residual Modeling
Input:
Actual AQI values y
Ensemble predictions y_pred_ensemble
Output:
Final hybrid AQI predictions y_final
Steps:

1. Compute residuals:
residuals = y - y_pred_ensemble

2. Determine optimal ARIMA parameters:
- Apply Auto-ARIMA using AIC to obtain (p, d, q).

3. Fit ARIMA model on residuals.

4. Predict residuals:
residual_pred = ARIMA.predict()

5. Obtain final hybrid prediction:
y_final = y_pred_ensemble + residual_pred

6. Evaluate performance using MSE, RMSE, and R^2.

7. Return final hybrid AQI predictions**.**

4.Algorithm: SHAP Analysis for AQI Prediction

Input:

- *Trained XGBoost model (Model_XGB)*
- *Feature dataset X (PM2.5, PM10, NO2, SO2, CO, O3)*
- *Target variable y (AQI)*

Output:

- *SHAP values for each feature*
- *Global and local feature importance visualizations*

Steps:

1. *Train the XGBoost model using the training dataset (X_train, y_train).*

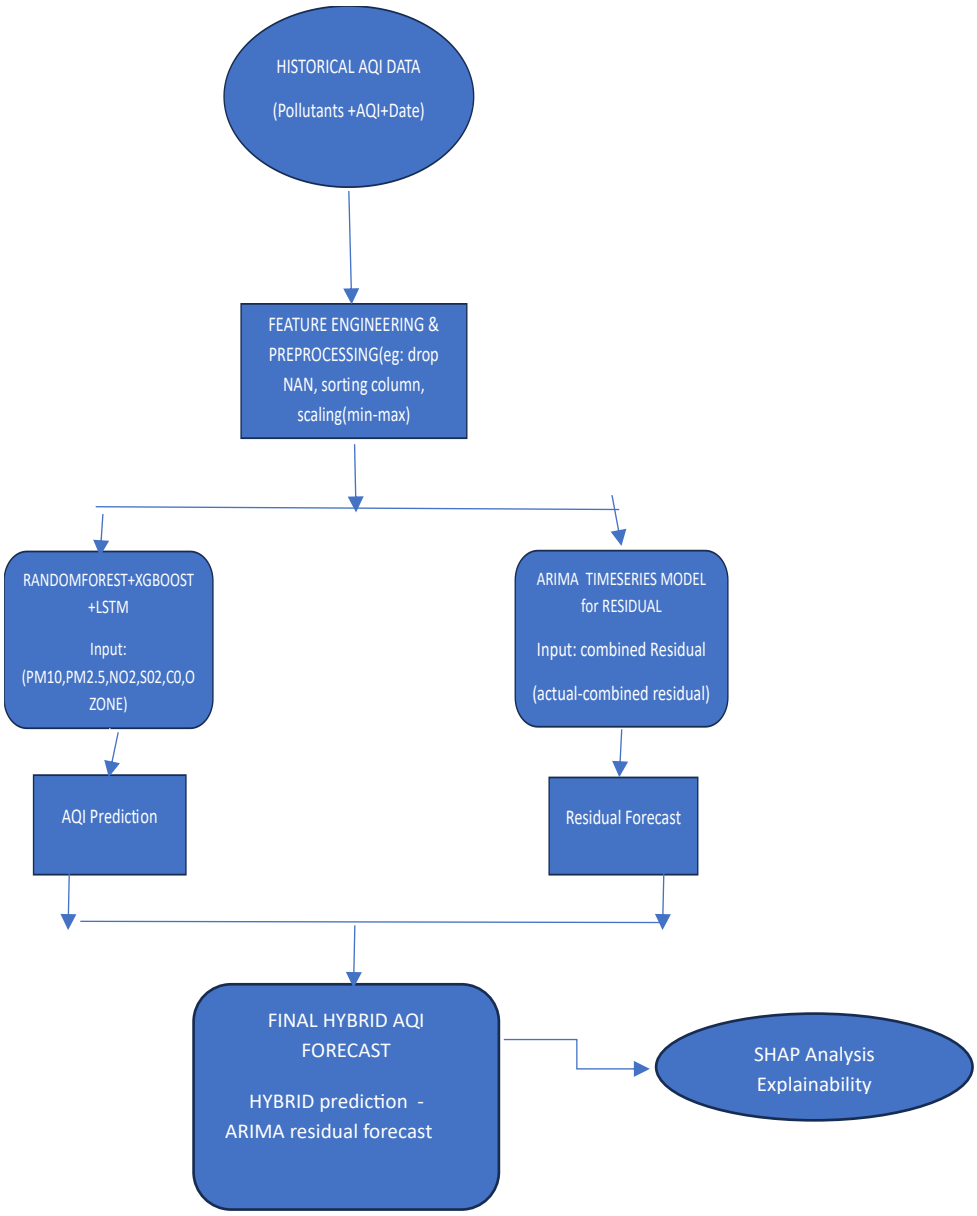
2. Initialize the SHAP TreeExplainer with the trained model.

3. Compute SHAP values for the test dataset to quantify feature contributions.

4. Generate global interpretations using SHAP summary plots and mean absolute SHAP values.

5. Produce local and dependence plots to explain individual predictions and feature effects on AQI.

5.2 FLOWCHART OF THE HYBRID (RFR+XGBOOST+LSTM) MODEL for AQI



5.3 MODEL DEVELOPMENT AND PERFORMANCE

I. Random Forest Regressor:

Random Forest is an ensemble learning method that constructs multiple decision trees and aggregates their predictions. It effectively captures nonlinear relationships between pollutants and AQI.

Mathematical Representation:

$$\hat{Y}_t^{RF} = \frac{1}{B}\sum_{b=1}^{B} T_b\,(X_t)$$

where $T_b$represents the b^{th}decision tree.

II. XGBoost (XGB)

XGBoost is a gradient boosting algorithm that builds trees sequentially, minimizing prediction errors using a regularized objective function. It offers high predictive accuracy and computational efficiency.

Objective Function:

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k=1}^{K} \Omega(f_k)$$

where $l$is the loss function and $\Omega$is the regularization term.

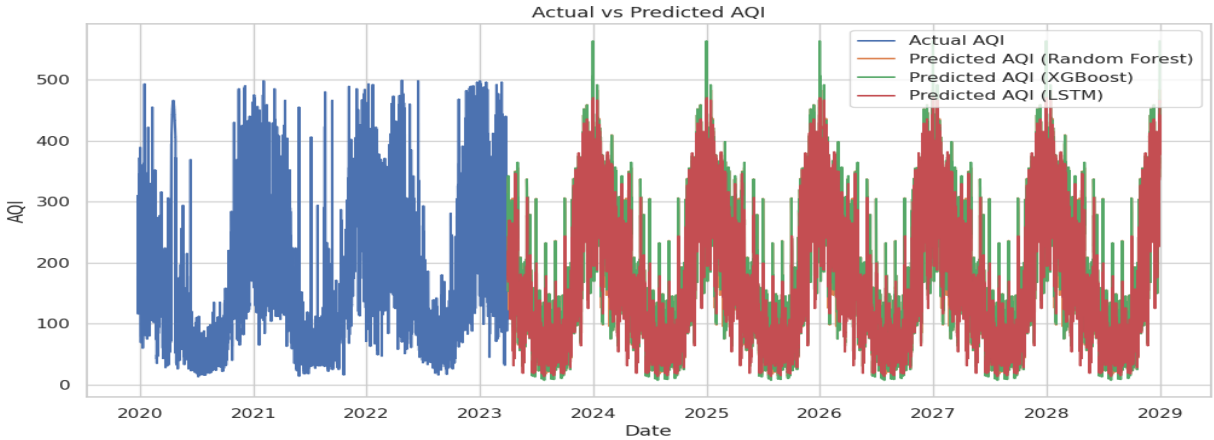
III. Long Short-Term Memory (LSTM)

LSTM networks are a type of recurrent neural network designed to capture long-term dependencies in sequential data. The gating mechanisms—input, forget, and output gates—enable the model to retain relevant temporal information.

Prediction Function:

$$\hat{Y}_t^{LSTM} = f_{LSTM}(X_{t-n}, \dots, X_t)$$

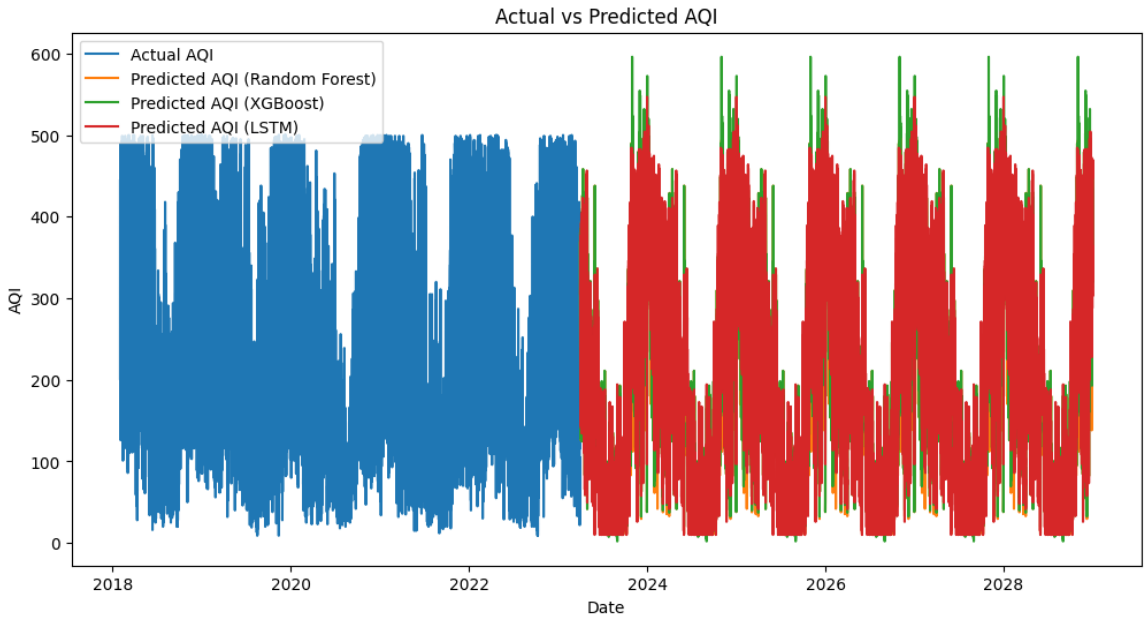
Fig 5.3.1: COMPARISON OF ACTUAL AQI TO PREDICTED (RFR,XGBOOST ,LSTM) FOR BR005:



INTERPRETATION:

Figure 5.3.1 : Represents a comparison between the actual AQI values (2020–mid 2023) and the predicted AQI generated by Random Forest, XGBoost, and LSTM models for the period 2023–2029. The models successfully capture the seasonal and cyclical behavior of air pollution, with XGBoost and LSTM showing closely aligned predictions and Random Forest providing slightly smoother estimates. The consistency among the models demonstrates the robustness of the forecasting framework and its suitability for long-term air quality assessment."

Fig 5.3.2: COMPARISON OF ACTUAL AQI TO PREDICTED (RFR,XGBOOST ,LSTM) FOR DL024:

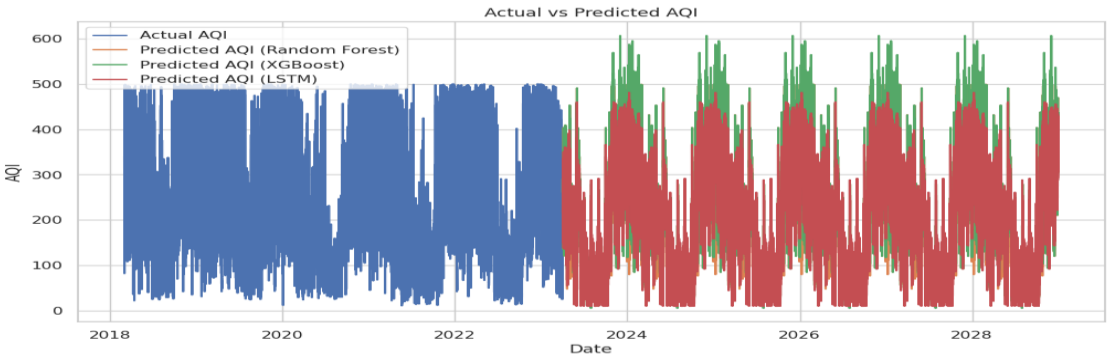


INTERPRETATION:

Fig 5.3.2:Represents observed AQI levels from 2018 to mid-2023, showing significant variability and multiple high pollution episodes. The predictions from the three models closely align with each other, indicating consistency and robustness in forecasting, with

XGBoost and LSTM capturing sharper peaks while Random Forest appears slightly smoother. Predicted values span from low to very high AQI levels (up to ~600), suggesting potential future pollution events.

Fig 5.3.3: COMPARISON OF ACTUAL AQI TO PREDICTED (RFR,XGBOOST ,LSTM) FOR DL030:

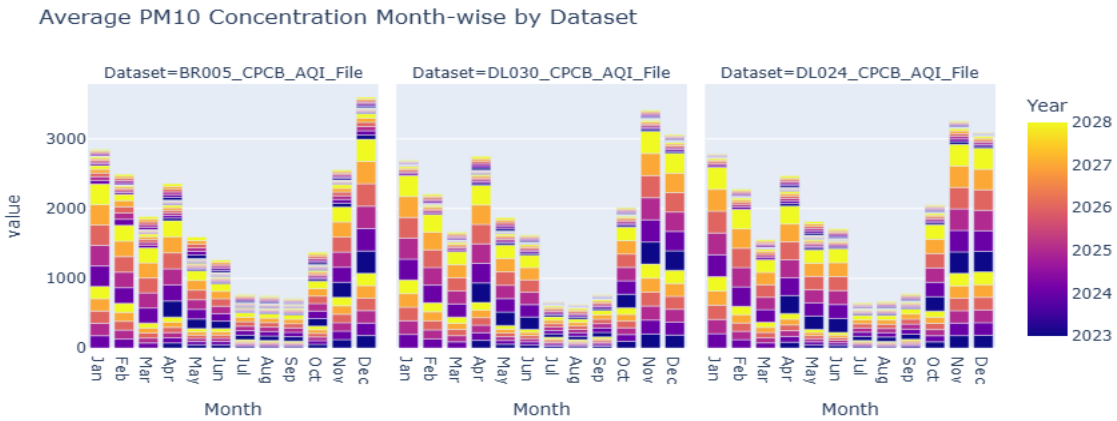


INTERPRETATION:

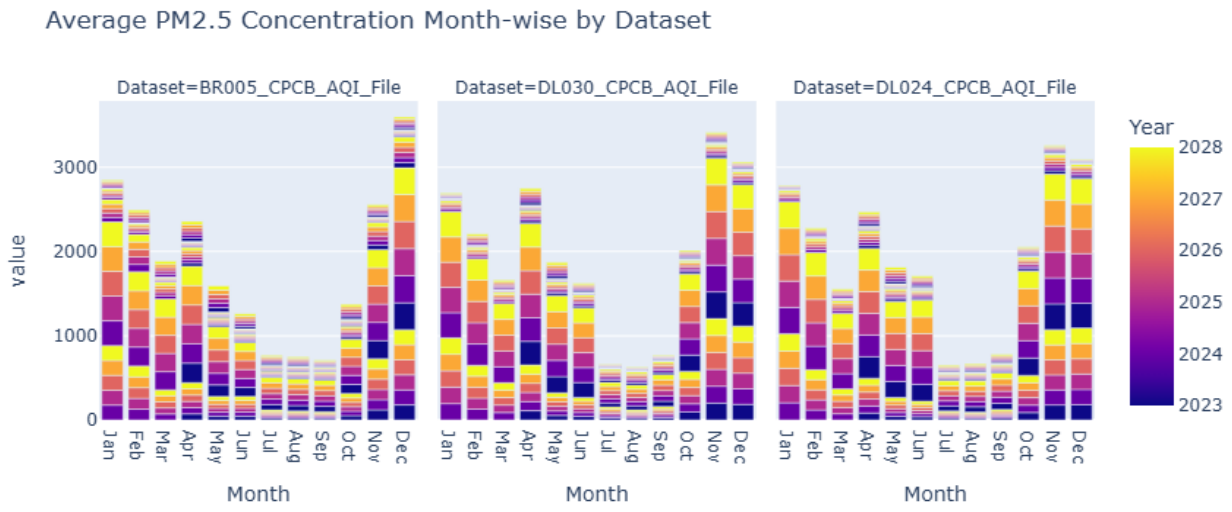
Fig 5.3.3: Represents observed AQI levels from 2018 to mid-2023. The data shows significant fluctuations with several high pollution episodes, indicating real-world variability in air quality. The Random Forest produces smoother predictions, capturing the general trend .Shows sharper peaks and higher AQI values ,indicating sensitivity to extreme pollution events.

5.3.2 POLLUTION AND ITS CONCENTRATION FOR ACTUAL DATASET AND PREDICTED DATASET:

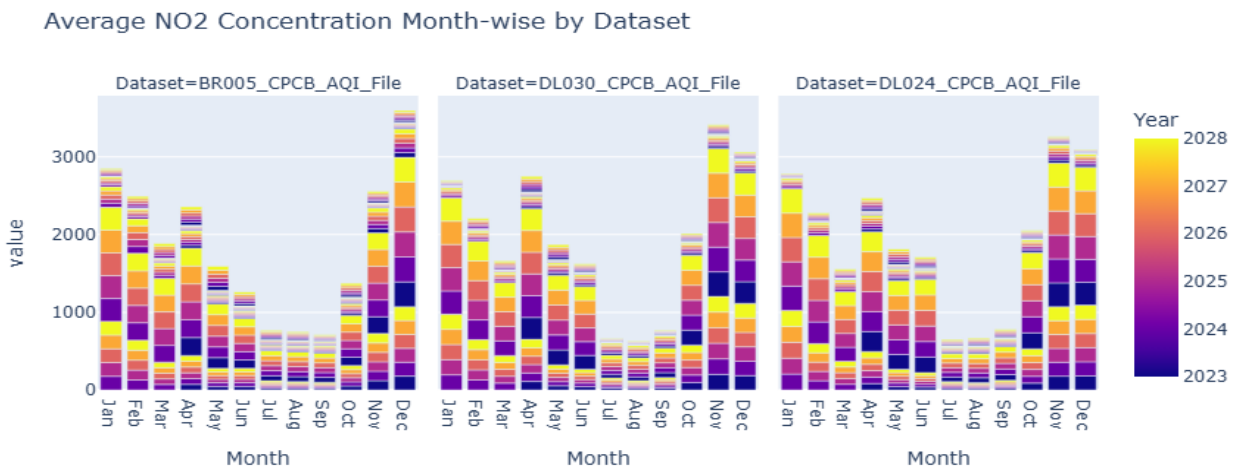
1. Comparison of PM10



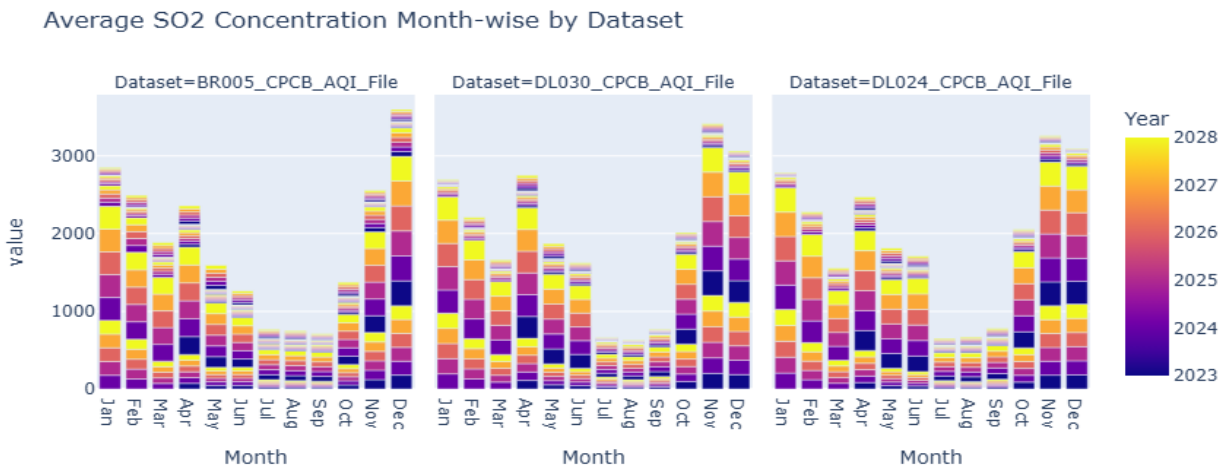
2. Comparison of PM2.5



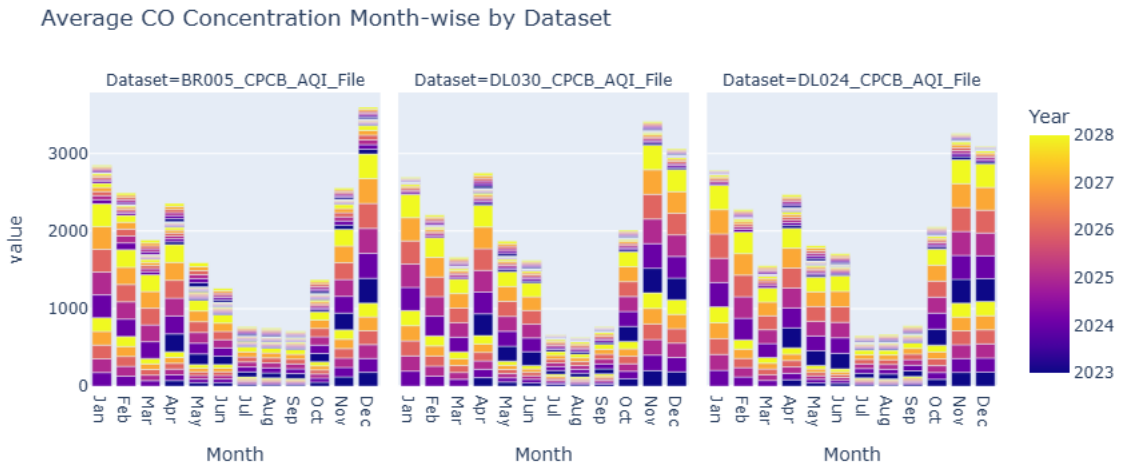
3.Comparison of NO2



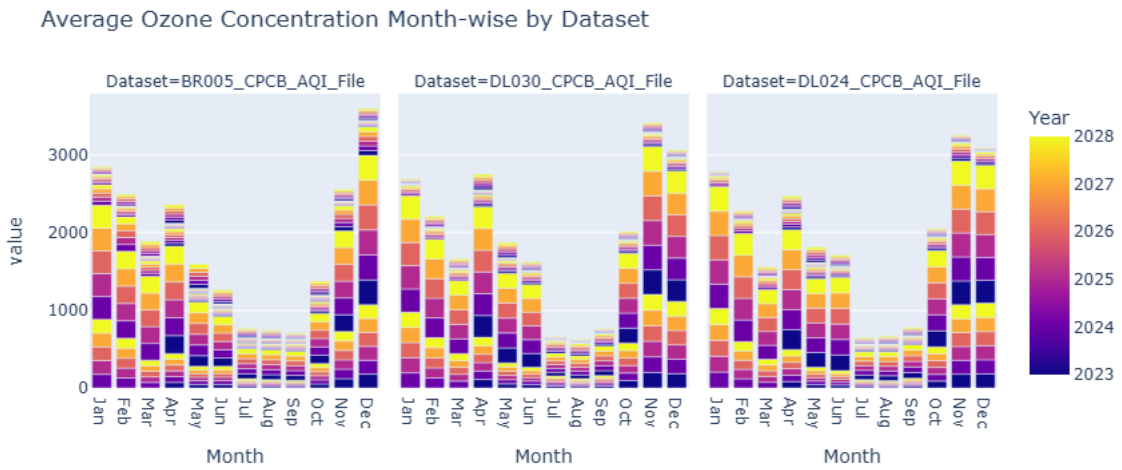
4. Comparison of SO2



5. Comparison of CO



6. Comparison of OZONE (DL030)



INTERPRETATION:

[1]A consistent seasonal trend is visible across all three datasets, showing that PM10 levels peak during the winter months (November to January) and drop to their lowest during the monsoon season (July to September).

[2]A consistent seasonal trend is visible across all three datasets, showing that PM2.5 levels peak during the winter months (November to January) and drop to their lowest during the monsoon season (July to September).

[3]All three datasets exhibit a consistent seasonal pattern, with NO2 concentrations peaking significantly during winter months (November to January) and reaching their lowest points during the monsoon season (July to September). the magnitude of concentration values varies, with noticeable peaks in November and December indicating heightened pollution levels during the end of each year.

[4]Concentrations of SO2 are highest during the winter months of November, December, and January, while they remain at their lowest during the monsoon period from July to September. The consistent increase in the height of the bars toward the end of the year suggests a recurring annual pattern of heightened sulfur-related emissions or atmospheric trapping.

[5]CO levels show a massive spike in December and November, whereas the levels drop significantly during the summer and rainy seasons. The data across BR005, DL030, and DL024 follows an almost identical trend, indicating that the drivers for CO pollution are likely uniform across these locations.

[6]Ozone typically shows secondary peaks in pre-monsoon months (like April/May) alongside the standard year-end highs seen in these specific visualizations. The low values from July to September across all years indicate that the monsoon season effectively cleanses or inhibits the formation of Ozone in these areas.

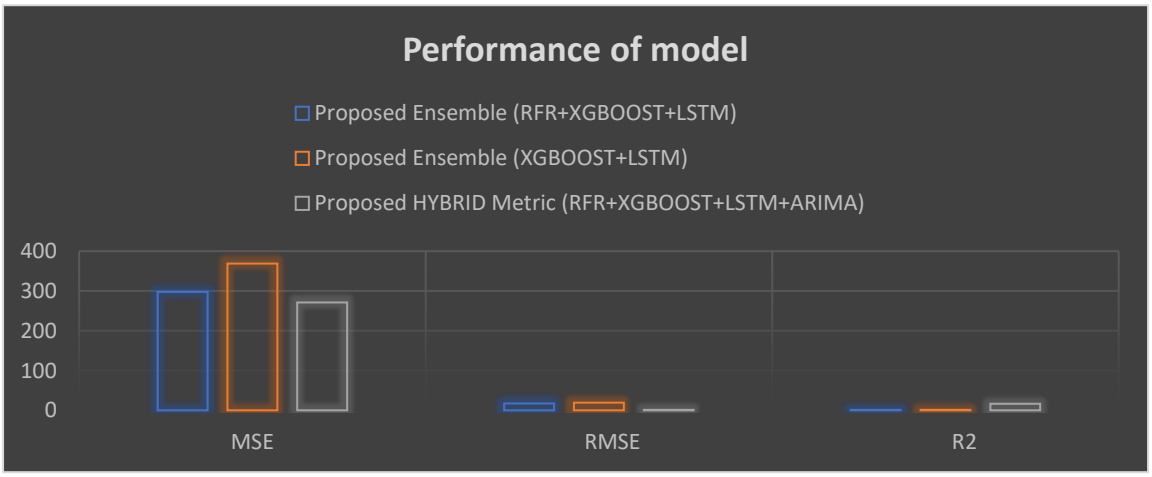
5.5 ENSEMBLE MODEL PERFORMANCE

Table I : EVALUATION METRIC FOR BR005:

A.

Proposed models for comparison metrics	MSE	RMSE	R^2
Proposed Ensemble (RFR+XGBOOST+LSTM)	297.86	17.26	0.98
Proposed Ensemble (XGBOOST+LSTM)	368.97	19.21	0.97
Proposed HYBRID Metric (RFR+XGBOOST+LSTM+ARIMA)	271.18	0.98	16.47

Fig I: comparison graph for best prediction model:



Ensemble (RFR + XGBoost + LSTM)

- **MSE:** 297.86
- **RMSE:** 17.26
- **R^2:** 0.98

Interpretation:

- Combining all three models significantly improves predictive accuracy. The high R^2 value (0.98) indicates that 98% of the variance in AQI is explained. The ensemble effectively reduces bias and variance compared to individual models.

B. Proposed Ensemble (XGBoost + LSTM)

- **MSE:** 368.97
- **RMSE:** 19.21
- **R^2:** 0.97

Interpretation:

- This model performs slightly worse than the full ensemble. The absence of the Random Forest component reduces the diversity of the ensemble. Although still strong, the higher MSE and RMSE indicate comparatively larger prediction errors.

C. Proposed Hybrid Model (RFR + XGBoost + LSTM + ARIMA)

- **MSE: 271.18**
- **RMSE: 16.47**
- **R^2: 0.98**

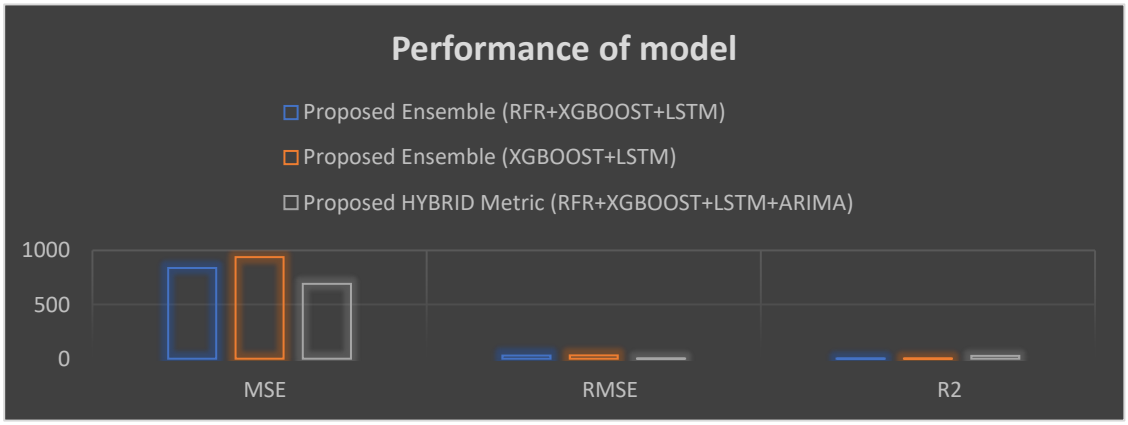
Interpretation:

- This is the **best-performing model** among the three. The **lowest MSE and RMSE** indicate superior prediction accuracy. The inclusion of **ARIMA-based residual modeling** captures remaining linear temporal patterns in the errors. Although the R^2 remains 0.98 (similar to the first ensemble), the reduction in error metrics confirms the effectiveness of the hybrid approach.

II. EVALUATION METRIC FOR DL024:

Proposed models for comparison metrics	MSE	RMSE	R^2
Proposed Ensemble (RFR+XGBOOST+LSTM)	838.45	28.96	0.95
Proposed Ensemble (XGBOOST+LSTM)	938.10	30.63	0.94
Proposed HYBRID Metric (RFR+XGBOOST+LSTM+ARIMA)	691.34	0.96	26.29

Fig II: comparison graph for best prediction model:



A. Proposed Ensemble (RFR + XGBoost + LSTM)

- **MSE: 838.45**
- **RMSE: 28.96**
- **R^2: 0.95**

Interpretation:
This ensemble leverages the strengths of tree-based models (RFR and XGBoost) and the temporal learning capability of LSTM. It provides strong predictive performance, explaining 95% of the variance in AQI values.

B. Proposed Ensemble (XGBoost + LSTM)

- **MSE: 938.10**
- **RMSE: 30.63**
- **R^2: 0.94**

Interpretation:
While still effective, this model performs slightly worse than the full ensemble. The absence of the Random Forest component reduces model diversity, leading to higher prediction errors and a marginally lower R^2.

C. Proposed Hybrid Model (RFR + XGBoost + LSTM + ARIMA)

- **MSE: 691.34**
- **RMSE: 26.29**
- **R^2: 0.96**

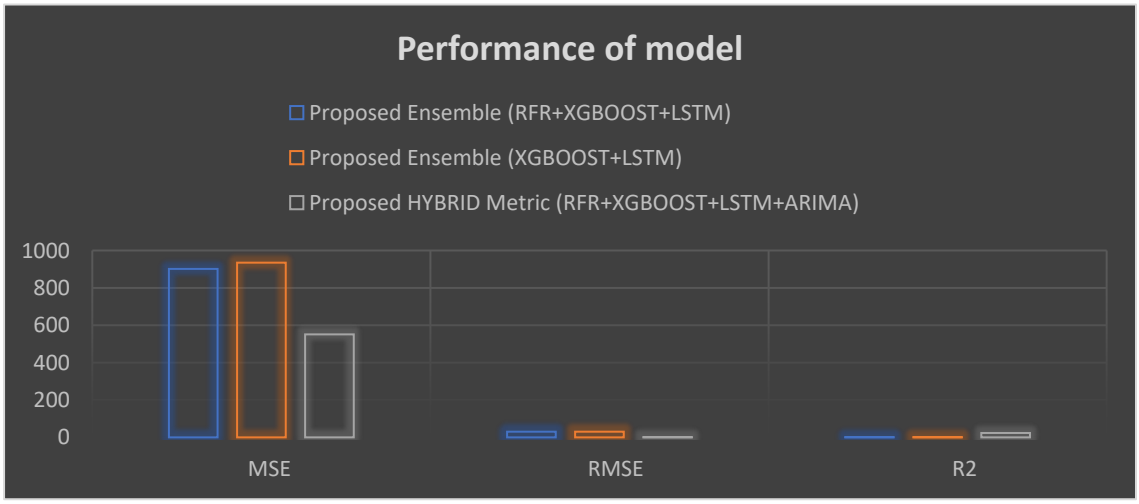
Interpretation:
This is the best-performing model among the three. By incorporating ARIMA-based residual modeling, the hybrid approach captures remaining linear temporal patterns in the ensemble's errors, resulting in:

- The lowest MSE and RMSE, indicating superior accuracy.
- The highest R^2, reflecting enhanced explanatory capability.

III. EVALUATION METRIC FOR DL030:

Proposed models for comparison metric	MSE	RMSE	R^2
Proposed Ensemble (RFR+XGBOOST+LSTM)	901.78	30.03	0.95
Proposed Ensemble (XGBOOST+LSTM)	934.80	30.57	0.95
Proposed HYBRID Metric (RFR+XGBOOST+LSTM+ARIMA)	551.57	0.97	23.48

Fig III: comparison graph for best prediction model:



A. Proposed Ensemble (RFR + XGBoost + LSTM)

- **MSE: 901.78**
- **RMSE: 30.03**
- **R^2: 0.95**

Interpretation:
This ensemble leverages the strengths of tree-based models (RFR and XGBoost) along with the temporal modeling capability of LSTM. It provides strong predictive performance, explaining 95% of the variance in AQI values.

B. Proposed Ensemble (XGBoost + LSTM)

- **MSE: 934.80**
- **RMSE: 30.57**
- **R^2: 0.95**

Interpretation:
Although this model still performs well, its accuracy is slightly lower than the full ensemble. The absence of the Random Forest component reduces model diversity, leading to marginally higher prediction errors.

C. Proposed Hybrid Model (RFR + XGBoost + LSTM + ARIMA)

- **MSE: 551.57**
- **RMSE: 23.48**
- **R^2: 0.97**

Interpretation:
This is the best-performing model among the three. By incorporating ARIMA-based residual modeling, the hybrid framework captures remaining linear temporal patterns in the ensemble's prediction errors. This results in:

- A substantial reduction in MSE and RMSE, indicating improved accuracy.
- An increase in R^2, demonstrating better explanatory capability.

BEST PERFORMAMCE MODEL IS :

Hybrid Model + ARIMA

↓ (Residual Error Correction)

Best Predictive Performance

5.5 RESIDUAL ANALYSIS

1. Ensemble Strategy

To leverage the strengths of individual models, their predictions are combined to form a single ensemble estimate. A simple average ensemble is employed:

$$\hat{Y}_t^{ENS} = \frac{\hat{Y}_t^{RF} + \hat{Y}_t^{XGB} + \hat{Y}_t^{LSTM}}{3}$$

Alernatively, a weighted ensemble may be used:

$$\hat{Y}_t^{ENS} = w_1\hat{Y}_t^{RF} + w_2\hat{Y}_t^{XGB} + w_3\hat{Y}_t^{LSTM}, \sum w_i = 1$$

2.Residual Calculation

The residuals represent the unexplained component of the ensemble prediction:

$$\boldsymbol{e_t = Y_t - \hat{Y}_t^{ENS}}$$

3. ARIMA Model

The **Auto-Regressive Integrated Moving Average (ARIMA)** model, denoted as ARIMA(p, d, q), is used to model residual patterns. The optimal parameters are selected using **Auto-ARIMA** based on information criteria such as AIC.

4 Final Hybrid Prediction

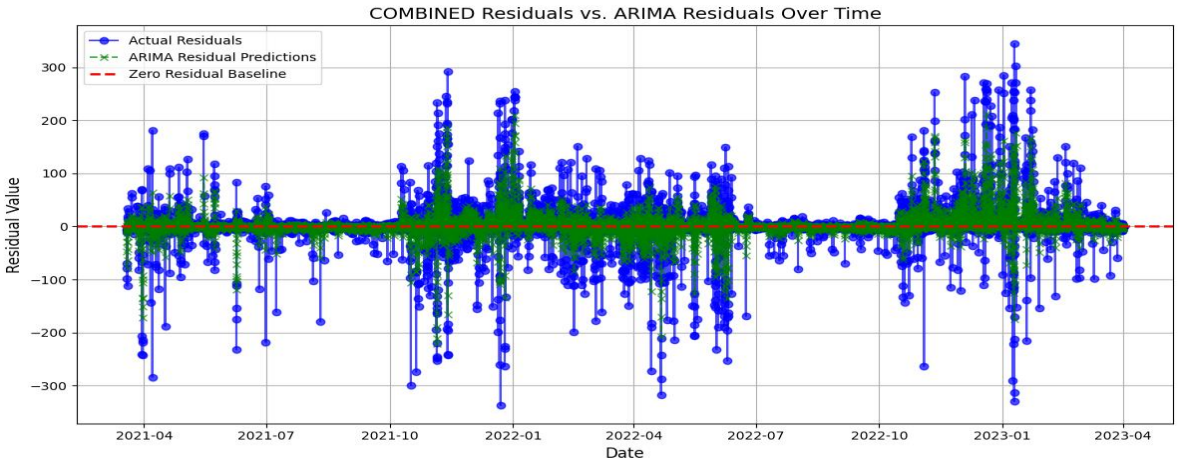
The final AQI forecast is obtained by adding the predicted residuals to the ensemble prediction:

$$\hat{Y}_t^{FINAL} = \hat{Y}_t^{ENS} + \hat{e}_t$$

$$\hat{Y}_t^{ENS} = AQI_{RFR} + AQI_{XGBOOST} + AQI_{LSTM}$$

RESULTS: From hybrid model like *rfr+ xgboost+ lstm+ ARIMA model* approach integrating this model via training and testing ,even with high-performing model ,some pattern in the residual (the differences between actual and predicted AQI) remained Unexplained. So the respective figures explains those errors .

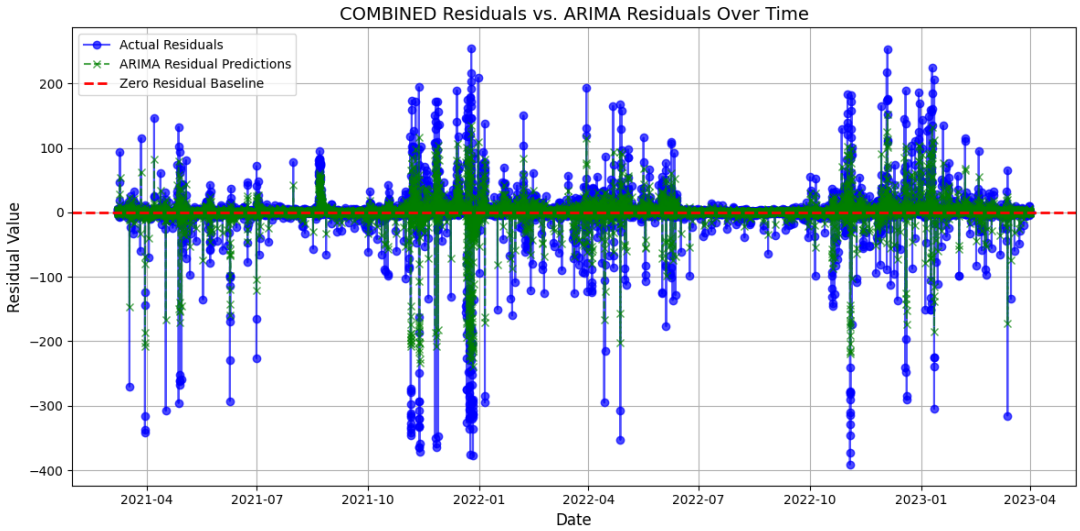
Fig 5.5.1: **RESIDUAL PLOT FOR TESTED COMBINED(ACTUAL) RESIDUAL VS ARIMA -Residual over time (DL030)**



INTERPRETATION:

The graph illustrates the comparison between the actual residuals of the ensemble AQI model and the residuals predicted by the ARIMA model over time. The actual residuals fluctuate around the zero -residual baseline, indicating that the ensemble model is generally unbiased but experiences occasional large errors during extreme pollution events. The ARIMA-predicted residuals closely follow the pattern of the actual residuals, its effectiveness in capturing remaining temporal dependencies in the prediction errors. Periods with higher scatterness of residuals suggest increased variability in air quality, likely influenced by seasonal factor. ARIMA-based residual modeling enhances the accuracy and reliability of the final hybrid AQI forecasts.

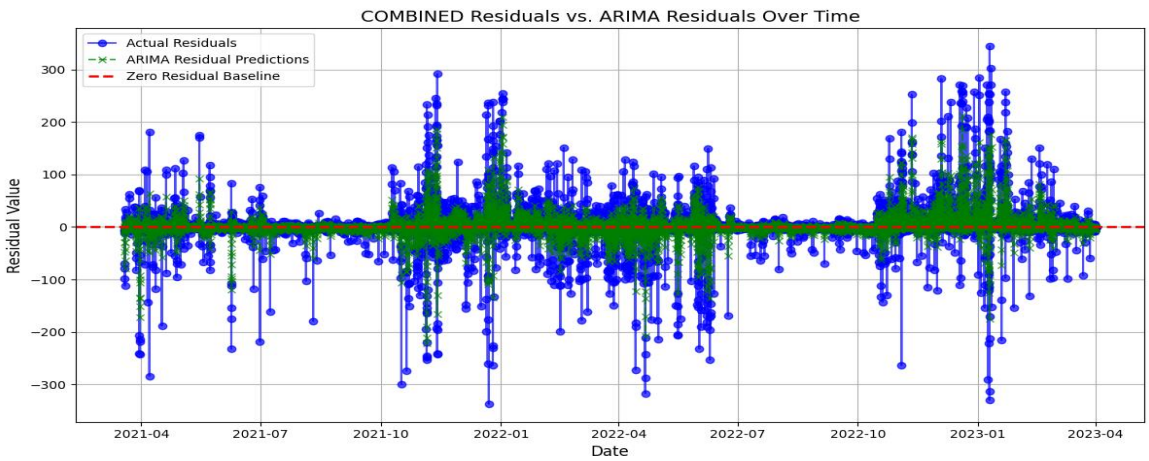
Fig 5.5.2: **RESIDUAL PLOT FOR TESTED COMBINED(ACTUAL) RESIDUAL VS ARIMA -Residual over time (DL024)**



INTERPRETATION:

the ensemble AQI model with the **ARIMA-predicted residuals** over time. The actual residuals fluctuate around the zero baseline, indicating that the ensemble model is largely unbiased but occasionally experiences large errors during extreme pollution events. The ARIMA residuals closely follow the pattern of the actual residuals, demonstrating its ability to capture remaining temporal dependencies in the prediction errors. Overall, the alignment between the two series confirms that ARIMA-based residual modeling enhances the accuracy of the final hybrid AQI forecasts.

Fig 5.5.3: **RESIDUAL PLOT FOR TESTED COMBINED(ACTUAL) RESIDUAL VS ARIMA -Residual over time (BR005)**



INTERPRETATION:

The ARIMA-predicted residuals closely follow the pattern of the actual residuals, its effectiveness in capturing remaining temporal dependencies in the prediction errors. Periods with higher scatterness of residuals suggest increased variability in air quality, likely influenced by seasonal factor. ARIMA-based residual modeling enhances the accuracy and reliability of the final hybrid AQI forecasts.

5.6 SHAP ANALYSIS FOR MODEL PERFORMANCE:

The SHAP value for a feature $i$is computed as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F|-|S|-1)!}{|F|!} \left[f_{S \cup \{i\}}(x) - f_S(x) \right]$$

Symbol	**Meaning**
(ϕ_i)	SHAP value (contribution) of feature (i)
(F)	Set of all features
(S)	Subset of features not containing (i)
(f_S(x))	Model prediction using feature subset (S)
(f_{S U{i}}(x))	Prediction after adding feature (i)
(	S
(	F

Despite its good performance, the hybrid model +ARIMA model is subject to some limitation. To start with hybrid model relies greatly on the presence and quality of historical pollutant concentration data; incomplete or inconsistent measurements may decrease prediction accuracy. However, the model designed is extensible. Exogenous variables such as meteorological parameters (e.g., temperature, wind speed, humidity), population density, and policy interventions can significantly influence air quality dynamics. They can be seamlessly integrated in future work, enhancing generalizability without modifying the core.

5.6.1 SHAP Summary:

1. SHAP PERFORMANCE OF DL024:

Rank	Pollutant	Most Impactful Feature	Feature Score (Mean Abs SHAP)	Reason
1	PM10 (ug/m3)	month	85.15240478515625	Monthly variations (seasonal patterns) are the most significant factor.
2	PM2.5 (ug/m3)	month	66.07247161865234	Monthly variations (seasonal patterns) are the most significant factor.
3	Ozone (ug/m3)	hour	14.912816047668457	Hourly patterns (e.g., traffic, industrial activity) are the most significant factor.
4	NO2 (ug/m3)	month	8.823256492614746	Monthly variations (seasonal patterns) are the most significant factor.
5	SO2 (ug/m3)	year	3.362262487411499	Yearly trends (long-term changes) are the most significant factor.
6	CO (mg/m3)	month	0.4128373861312866	Monthly variations (seasonal patterns) are the most significant factor.

2. SHAP PERFOMANCE OF DL030:

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Rank	Pollutant	Most Impactful Feature	Feature Score (Mean Abs SHAP)	Reason
1	PM10 (ug/m3)	month	83.0460205078125	Monthly variations (seasonal patterns) are the most significant factor.
2	PM2.5 (ug/m3)	month	56.52363586425781	Monthly variations (seasonal patterns) are the most significant factor.
3	Ozone (ug/m3)	hour	20.31132698059082	Hourly patterns (e.g., traffic, industrial activity) are the most significant factor.
4	NO2 (ug/m3)	month	10.239503860473633	Monthly variations (seasonal patterns) are the most significant factor.
5	SO2 (ug/m3)	year	5.427920341491699	Yearly trends (long-term changes) are the most significant factor.
6	CO (mg/m3)	hour	0.4534723460674286	Hourly patterns (e.g., traffic, industrial activity) are the most significant factor.

3. SHAP PERFORMANCE OF BR005:

index	Rank	Pollutant	Most Impactful Feature ▲	Feature Score (Mean Abs SHAP)	Reason
3	4	Ozone (ug/m3)	hour	12.988945960998535	Hourly patterns (e.g., traffic, industrial activity) are the most significant factor.
0	1	PM10 (ug/m3)	month	73.98988342285156	Monthly variations (seasonal patterns) are the most significant factor.
1	2	PM2.5 (ug/m3)	month	42.78194808959961	Monthly variations (seasonal patterns) are the most significant factor.
2	3	NO2 (ug/m3)	month	23.807598114013672	Monthly variations (seasonal patterns) are the most significant factor.
4	5	SO2 (ug/m3)	month	2.929384708404541	Monthly variations (seasonal patterns) are the most significant factor.
5	6	CO (mg/m3)	month	0.441470742225647	Monthly variations (seasonal patterns) are the most significant factor.

5.6.2 Interpretation of Pollutant Contributions

In the model, **PM2.5 and PM10** typically exhibit the highest SHAP values, indicating that they are the most influential pollutants driving AQI levels due to their strong association with particulate pollution. **NO_2** also shows a significant positive contribution, reflecting its link with vehicular emissions and urban activities. **O_3 (Ozone)** demonstrates a variable impact, sometimes increasing AQI depending on photochemical conditions, especially during high sunlight periods.

On the other hand, **SO_2 and CO** generally show relatively lower SHAP contributions, suggesting a smaller influence on AQI compared to particulate matter and nitrogen oxides in the given dataset. The SHAP summary and dependence plots further reveal that higher concentrations of key pollutants, particularly PM2.5, consistently lead to a sharp increase in AQI predictions.

Overall, SHAP analysis confirms that particulate matter (PM2.5 and PM10) are the dominant drivers of air pollution in the model, while other pollutants contribute to variations depending on environmental and seasonal conditions, thereby enhancing the interpretability and reliability of the AQI prediction framework.

CONCLUSION

This study presents a highly accurate and understandable system for forecasting air quality by thoughtfully combining several machine learning techniques. To begin, the researchers had to ensure their foundation was solid. Missing data can easily derail predictive models, so they used linear interpolation to fill in any gaps, keeping the historical timeline smooth and reliable. They also calculated the Air Quality Index (AQI) using official CPCB guidelines, which was a necessary step to ensure their final predictions would be standardized and immediately useful to government agencies and the public.

Once the data was prepped, they tackled the actual forecasting by blending Random Forest, XGBoost, and LSTM models into a single "ensemble." This specific combination was chosen for a very practical reason: air pollution is incredibly complex. The tree-based models (XGBoost and Random Forest) were deployed to untangle the complicated, non-linear ways different pollutants interact with each other, while the LSTM model was brought in specifically for its ability to remember "temporal dynamics," or how pollution naturally rises and falls over days and seasons. By averaging these models together, the team successfully smoothed out the individual biases and weaknesses of each, resulting in a much more robust prediction.

However, even the best AI leaves behind tiny errors. To squeeze out every last drop of accuracy, the researchers took the clever step of using an ARIMA model to analyze the ensemble's leftover "residual errors." This was done to catch any lingering time-based patterns the main AI might have missed, ultimately refining the forecast and proving its success through strong evaluation metrics like MSE, RMSE, and R^2.

Finally, because machine learning often acts like a mysterious "black box," the team used SHAP analysis to transparently explain exactly how the model reached its conclusions. This was a critical step for building real-world trust, as it allowed researchers to quantify exactly which pollutants were driving the predictions. It revealed that particulate matter (PM2.5 and PM10) are the dominant forces worsening air quality, far outweighing gases like SO2 and CO. Ultimately, by carefully layering these techniques—from data smoothing to error correction and transparent AI logic—this hybrid model provides a highly reliable, actionable tool that policymakers can confidently use to design better pollution controls and protect public health.

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